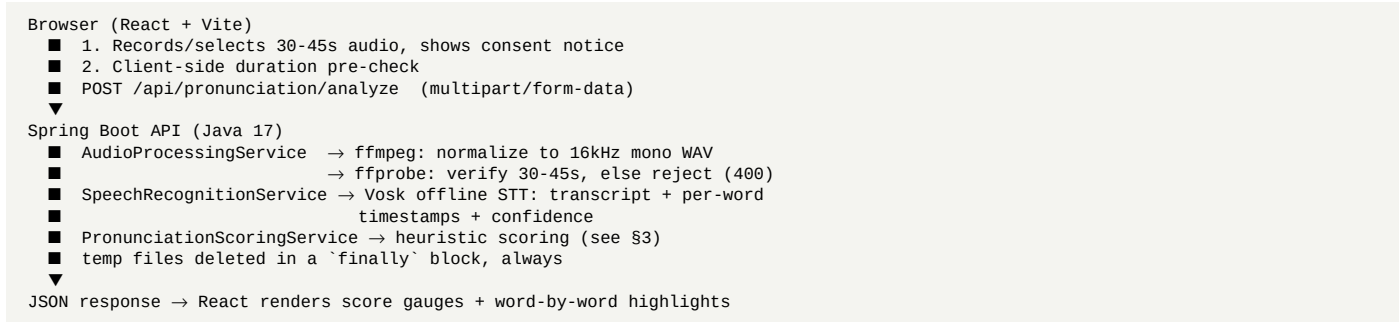


System Architecture — Livo AI Pronunciation Coach

1. Components



Frontend and backend are two independently deployable apps talking over HTTPS/CORS — no shared session state, no server-rendered pages.

2. Models & APIs used, and why

Need	Choice	Why over the alternative
Speech-to-text w/ word timing & confidence	Vosk (small English model, vosk-model-small-en-us-0.15)	Free, Apache-2.0, runs fully offline/on our own server — no API key, no per-request billing, no audio leaving our infrastructure to a third party. The brief explicitly ruled out paid keys, which rules out SpeechSuper/Azure Pronunciation Assessment/Google Cloud Speech for this deliverable.
Audio normalization	ffmpeg/ffprobe	Free, industry-standard, handles arbitrary input codecs/containers and gives exact duration for the 30-45s constraint.
Backend framework	Spring Boot 3	Matches the JD directly; mature multipart handling, easy to containerize and deploy to any free host.
Frontend	React + Vite	Matches the JD; Vite gives fast dev/build with zero-config static output deployable anywhere.

3. How pronunciation is scored, and what gets highlighted

There is no paid phoneme-level pronunciation-assessment API in this build, so scoring is built from signals a speech recognizer already gives us for free:

- **Per-word confidence** (0–1) from Vosk is used as a proxy for clarity: a recognizer trained on native English speech tends to lose confidence on words that are mispronounced, slurred, or acoustically ambiguous — the same words a human listener would likely stumble on.
 - confidence $\geq 0.80 \rightarrow$ **good**
 - $0.55 \leq \text{confidence} < 0.80 \rightarrow$ **unclear** ("said indistinctly")
 - confidence $< 0.55 \rightarrow$ **mispronounced** (didn't confidently match an expected English sound)
 - **Fluency** combines speaking rate (ideal band: 110–170 wpm) and a penalty for unnaturally long gaps between words ($> 0.8\text{s}$), which usually indicate hesitation or filler pauses.
 - **Overall score** $= 0.7 \times \text{clarity} + 0.3 \times \text{fluency}$, 0–100.
- The frontend highlights each word inline (amber = unclear, red = likely mispronounced), and a "words to practice" table lists every flagged word with its score and reason — this is the part a real learner would actually use.

Honest limitation: this is a confidence-based heuristic, not true phoneme-level scoring (comparing the learner's phonemes against a reference pronunciation, as SpeechSuper/Azure do). It correctly flags words that are hard to understand, but can't yet distinguish "wrong vowel sound" from "correct word, just quiet." See §5.

4. DPDP Compliance (Digital Personal Data Protection Act, 2023)

Voice recordings are personal data under DPDP. This build treats them as sensitive and minimizes retention rather than trying to bolt on compliance after the fact:

- **Consent:** the frontend shows an explicit consent notice before the upload control is enabled; the file picker is disabled until the user checks it. Uploading is the consent action (Section 6, DPDP).

- **Storage:** the API is stateless. Uploaded audio is written to a request-scoped OS temp file, converted, scored, and the temp files are deleted in a `finally` block immediately after the response is built — in the same request, regardless of success or failure. No database, no object storage, no logging of audio bytes or transcripts.

- **Retention:** effectively zero. Nothing outlives the HTTP request that created it.

- **Data residency:** the backend is a single container with no external STT API call (Vosk runs in-process), so audio never crosses a border to a third-party processor. Deploying the container to an India-region host (e.g., a Mumbai/Bangalore region on Render/Railway/Fly.io) keeps processing entirely within India, satisfying data-residency-conscious deployments without any code change.

- **Deletion:** there is nothing to delete after the response returns — by design, rather than by a deletion request workflow. If a future version adds accounts/history, deletion would need an explicit right-to-erasure endpoint per DPDP Section 12.

- **Purpose limitation:** audio is used only to produce the pronunciation score in that request; it is never used to train or fine-tune any model.

5. Trade-offs, and what's next with another week

Trade-offs made for this scope:

- Confidence-based scoring instead of true phoneme alignment (fast to build, free, but less precise than a purpose-built pronunciation model).
- Small Vosk model (~40MB) for fast cold-starts on free hosting tiers, trading some transcription accuracy for deploy simplicity.
- No user accounts/history — every analysis is a one-off, which keeps the DPDP story simple but means no progress tracking.

With another week:

1. Real phoneme-level scoring: align recognized phonemes (forced-aligner or `whisper + G2P`) against expected pronunciation (CMUdict) instead of relying purely on recognizer confidence.
2. Swap in the larger Vosk model (or on-prem Whisper) behind a feature flag for accuracy-sensitive deployments.
3. A short reference-text (read-aloud) mode so scoring can compare against a known target sentence, not just open speech.
4. Optional accounts with a real deletion/export endpoint if history tracking is wanted, with explicit per-DPDP data-principal rights.
5. Rate limiting and true file-type sniffing (not just extension) at the API boundary for hardening.